

Theoretical Outline

This section describes the equations and variables used in GEM and the approach for solving these equations. The flow equations are discretized using the adaptive-implicit method ([Collins et al.,\(1992\)](#), [Thomas and Thurnau,\(1983\)](#), [Nghiem and Li,\(1989\)](#)), because i encompasses both the explicit-transmissibility method and the fully-implicit method as particular cases.

The equations, variables and solution method presented in the following are variations of the approach of [Collins et al.,\(1992\)](#).

Flow Equations

The material-balance finite-difference equations for the components in the oil and gas phases, and for the water component are:

$$\begin{aligned} \psi_i \equiv & \Delta T_o^m y_{io}^m (\Delta p^{n+1} - \gamma_o^m \Delta D) + \Delta T_g^m y_{ig}^m (\Delta p^{n+1} + \Delta P_{cog}^m - \gamma_g^m \Delta D) \\ & + q_i^m - \frac{V}{\Delta t} [N_i^{n+1} - N_i^n] = 0 \quad i = 1, \dots, n_c \end{aligned} \quad (A1.1)$$

$$\psi_{n_c+1} \equiv T_w^m (\Delta p^{n+1} - \Delta P_{cwo}^m - \gamma_w^m \Delta D) + q_{n_c+1}^m - \frac{V}{\Delta t} [N_{n_c+1}^{n+1} - N_{n_c+1}^n] = 0 \quad (A1.2)$$

where $N_i (i = 1, \dots, n_c)$ denote the moles of Component i per unit of gridblock volume, and where N_{n_c+1} denotes the moles of water per unit of gridblock volume. All other symbols are defined in the nomenclature. It is assumed that no mass transfer exists between the hydrocarbon and water phases. The superscripts n and $n+1$ denote respectively the old and current time level. The superscript m refers to n for explicit gridblocks and $n+1$ for fully-implicit gridblocks. In GEM, the term explicit refers to gridblocks with explicit transmissibilities where only pressure is treated implicitly.

The N_i 's are related to porosities phase molar densities, saturations and compositions as follows:

$$N_i = \phi (\rho_o S_o y_{io} + \rho_g S_g y_{ig}) \quad i = 1, \dots, n_c \quad (A1.3)$$

$$N_{n_c+1} = \phi \rho_w S_w \quad (\text{A1.4})$$

Phase-Equilibrium Equations

If the hydrocarbon system is in the two phase region at a given p , T and $N_i (i = 1, \dots, n_c)$, the phase compositions and splits can be obtained by solving the thermodynamic-equilibrium equation

$$\mathbf{g}_i \equiv \ln f_{ig} - \ln f_{io} = 0 \quad i = 1, \dots, n_c \quad (\text{A1.5})$$

for N_{ig} , the moles of Component i in the gas phase. The moles of Component i in the oil phase, N_{io} , can be obtained from

$$N_{io} = N_i - N_{ig} \quad i = 1, \dots, n_c \quad (\text{A1.6})$$

Saturation Equation

The saturations are related to N_i and $\rho_m (m = o, g, w)$ through the following equation

$$S_w = N_{n_c+1} / (\phi \rho_w) \quad (\text{A1.7})$$

$$S_o = (1 - S_w) \frac{N_o / \rho_o}{N_o / \rho_o + N_g / \rho_g} \quad (\text{A1.8})$$

$$S_g = (1 - S_w) \frac{N_g / \rho_g}{N_o / \rho_o + N_g / \rho_g} = 1 - S_w - S_o \quad (\text{A1.9})$$

Mole or Volume Consistency Equations

From the definition of $N_i (i = 1, \dots, n_c+1)$, one can write

$$\psi_p \equiv \sum_{i=1}^{n_c+1} N_i^{n+1} - \phi^{n+1} (\rho_o S_o + \rho_g S_g + \rho_w S_w)^{n+1} = 0 \quad (\text{A1.10})$$

Equation (A1.10) forces the consistency between the N_i 's and the densities, saturations, and porosities.

Equation (A1.10) can also be rewritten as follows

$$\psi'_p \equiv V \frac{\sum_{i=1}^{n_c+1} N_i^{n+1}}{(\rho_o S_o + \rho_g S_g + \rho_w S_w)^{n+1}} - V \phi^{n+1} = 0 \quad (\text{A1.11})$$

The first term in Equation (A1.11) is the volume occupied by the fluids (oil, gas, water) and the second term is the pore volume. Thus Equation (A1.11) forces the consistency between the fluid volume and the pore volume. A similar equation is referred to as volume balance equation by [Acs et al.,\(1985\)](#) and [Watts \(1986\)](#).

Decoupled Flash-Calculation Approach

Equations [\(A1.11\)](#), [\(A1.1\)](#), [\(A1.2\)](#), and [\(A1.5\)](#) form a system of $n_b(2n_c+2)$ non-linear equation which can, in principal, be solved for the variables

$$(p, N_i, \dots, N_{n_c}, N_{n_c+1}, N_{1g}, \dots, N_{n_{cg}})^{n+1}_k \quad k = 1, \dots, n_b$$

where n_b is the number of gridblocks. This approach will be referred to as the simultaneous-solution approach.

To alleviate the complexity associated with the simultaneous-solution of all governing equations, [Collins et al.,\(1992\)](#) decoupled the phase-equilibrium equations and solved them in an inner loop. The technique used in GEM is a variation of their approach and will be referred to as the decoupled-flash-calculation approach. The equations and primary variables of gridblock k are

$$\mathbf{F}_k = (\psi'_p, \psi_1, \dots, \psi_{n_c+1})^T_k \quad k = 1, \dots, n_b \quad (\text{A1.12})$$

$$\mathbf{X}_k = (p^{n+1}, N_1^{n+1}, \dots, N_{n_c+1}^{n+1})_k^T \quad k = 1, \dots, n_b \quad (\text{A1.13})$$

After every Newtonian iteration of $\mathbf{X}$, the phase-equilibrium Equation [\(A1.5\)](#) are solved to convergence.

The approach taken here separates the task of solving the consistency and flow equations from that of solving the phase-equilibrium equations. This is a desirable feature because flash calculations are localized in a module that can be developed independently. It also allows the use of different calculation techniques (e.g. Newton's method, quasi-Newton method, QNSS) and the easy implementation of stability-test procedures.

It is possible that this decoupled-flash-calculation approach may not be as efficient as the simultaneous-solution approach for systems with simple phase behavior, but it provides additional flexibility in implementing more robust flash algorithms for complex systems. It also allows the ability to repeat the flash calculations for gridblocks where convergence to unrealistic or metastable phases are obtained. The simultaneous-solution approach may require a repeat of the timestep. Note that the decoupled flash-calculation approach enforces the thermodynamic equilibrium between oil and gas at every iteration whereas in the simultaneous-solution approach, equilibrium is only satisfied after the convergence of the timestep.

Nomenclature

D	depth
f_{ij}	fugacity of component i in phase j
F	function
$\mathbf{g}$	phase-equilibrium function
n_b	number of gridblocks
n_c	number of components
N_i	moles of component i per unit block volume
p	pressure
P_{cog}	oil-gas capillary pressure
P_{cwo}	water-oil capillary pressure
q	injection/production rate
t	time
T_j	transmissibility of phase j
V	gridblock volume
y_{ij}	mole fraction of component i in phase j
γ	specific gravity or gravity term in flow equation
Δt	timestep
ρ_m	molar density of phase m
ϕ	porosity
ψ	function

Superscripts

(k) iteration level

n old time level

$n+1$ new time level

Subscripts

i component

j phase

o oil

g gas

w water